

# Developing a Crypto Trading Strategy Leveraging On-Chain Data

Togay Atmaca – 12268957

Jordan Cassella – 12452926

Matthew Haimes – 12452937

Kunj Shah – 12258505

Yan Song Zhao – 12448802

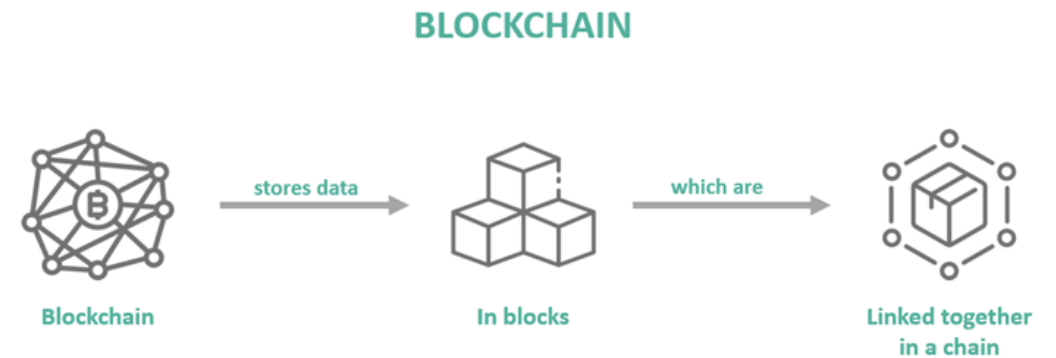
# Agenda

| Overview                                                                                                                                                              | Strategy Overview                                                                                                                                | MLP Model                                                                                                                                       | Analysis                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Blockchain Discussion</li><li>• Proof of Work (PoW) vs. Proof of Stake (PoS)</li><li>• Correlation of Token Returns</li></ul> | <ul style="list-style-type: none"><li>• Motivation</li><li>• Investment Universe</li><li>• Feature Space</li><li>• Strategy Discussion</li></ul> | <ul style="list-style-type: none"><li>• Parameterization</li><li>• Segmentation</li><li>• Model Construction</li><li>• Model Pipeline</li></ul> | <ul style="list-style-type: none"><li>• Results</li><li>• Portfolio Allocation</li><li>• Risk Management</li><li>• Conclusion</li></ul> |



# Blockchain

- Ledger records transactions securely across multiple computers.
- Distributed storage keeps copies across a network of nodes.
- Blocks bundle transactions and update the ledger.
- Blockchain ensures decentralization, security, and immutability.
- **Proof of Stake (PoS)** is part of the **consensus mechanism**, which determines how new blocks are added to the blockchain



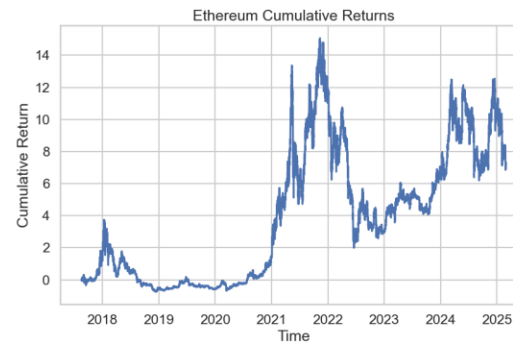
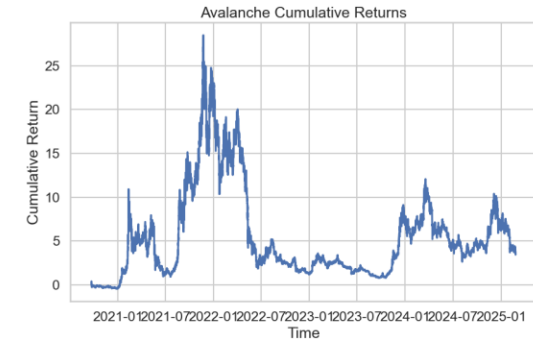
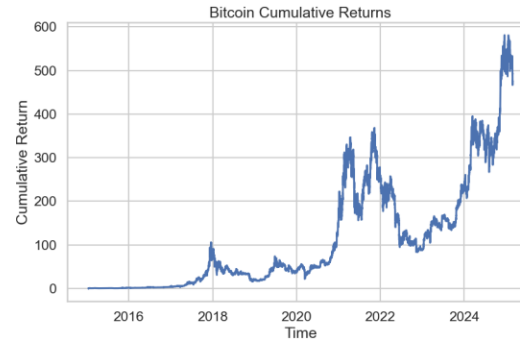
# Proof of Work (PoW)

- PoW is a consensus mechanism in the blockchain network to validate transactions, by miners solving complex problems
  - This takes significant computation power. The winner adds a new block on to blockchain. This is very energy intensive

# Proof of Stake (PoS)

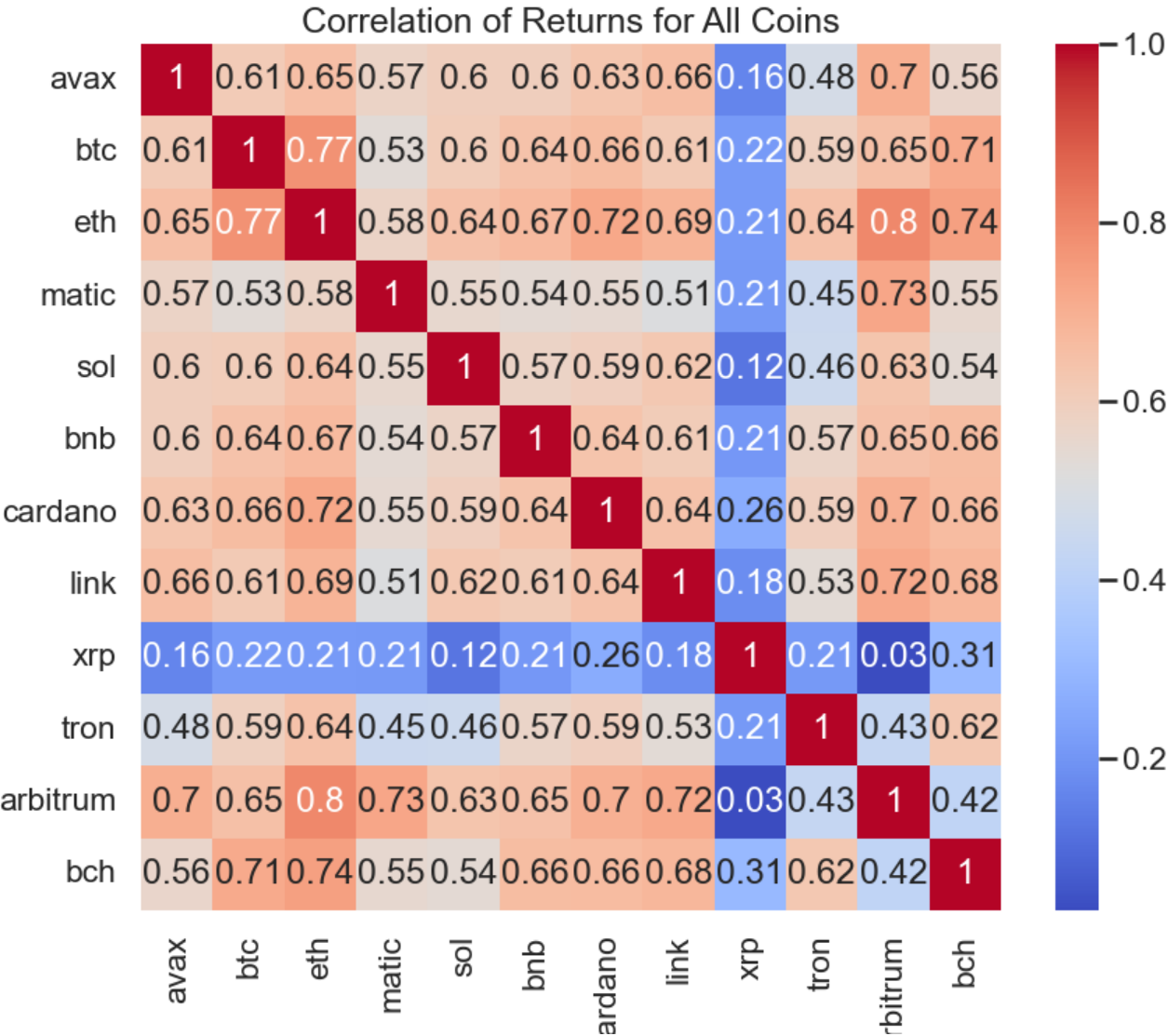
- PoS accomplishes the same without requiring high computational power.
  - Instead of miners, the PoS uses validators to lock up a certain amount of cryptocurrency as a stake
  - The validators create new blocks based on the amount staked and other factors like staking duration
  - The more stake the validator has increases the chance of being selected
  - More energy efficient

# Proof of Work (PoW) vs Proof of Stake (PoS) Tokens



- Bitcoin is one of the last remaining PoW tokens in the market, with many others (e.g. Ethereum) switching to PoS in the past 5 years. Thus, Bitcoin will be our main benchmark for PoW token performance.

# Correlation of Token Returns



We observe instances of uncorrelated returns between our tokens of interest. We interpret this as a positive signal for diversification in our strategy's ultimate portfolio allocation.

# Motivation

- Each asset class relies on fundamental data to determine proper valuation
- Traditional prediction models for cryptocurrency are limited by focusing on either:
  - Technical indicators from price history (Parente et al., 2024)
  - On-chain fundamental data (Kim et al., 2022)
- Our model bridges this gap by combining both data types and various methodologies from both papers
- This integrated approach delivers more accurate buy/hold/sell signals for enhanced risk management and portfolio allocation

| Fundamental Data Types |                  |                     |                 |
|------------------------|------------------|---------------------|-----------------|
| Equities               | Fixed Income     | Commodities         | Crypto          |
| - Financial Statements | - Interest Rates | - Supply and Demand | - On Chain Data |



# Hypothesis and Economic Rationale

- On-chain data metrics are fundamentally related to the price of PoW coins due to the nature of the block processing system
- Processing blocks in PoS systems is fundamentally related to how much validators are staking
- We hypothesize that because PoS on-chain metrics have a fundamental, contemporaneous relationship to asset price, there would not be as much predictive power looking into the future
- Therefore, we believe PoS on-chain metrics have a lead-lag relationship to digital asset price, where the information contained in the on-chain metrics are not instantaneously reflected in digital asset price, but instead, it's leading the asset price
- As a result, this lead-lag relationship can leave us a window to predict price directions in a profitable quantitative trading strategy

# Investment Universe

|     | Investment Securities                                                                                                                                              | Data                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| POS | <ul style="list-style-type: none"><li>- Avalanche(AVAX)</li><li>- Polka dot (DOT)</li><li>- Near (NEAR)</li><li>- Polygon (MATIC)</li><li>- Solana (SOL)</li></ul> | <ul style="list-style-type: none"><li>- Technical Time Series</li><li>- On Chain Block Data</li><li>- Sourced from Dune API</li></ul> |

# Examples of Feature Space for On-Chain Data

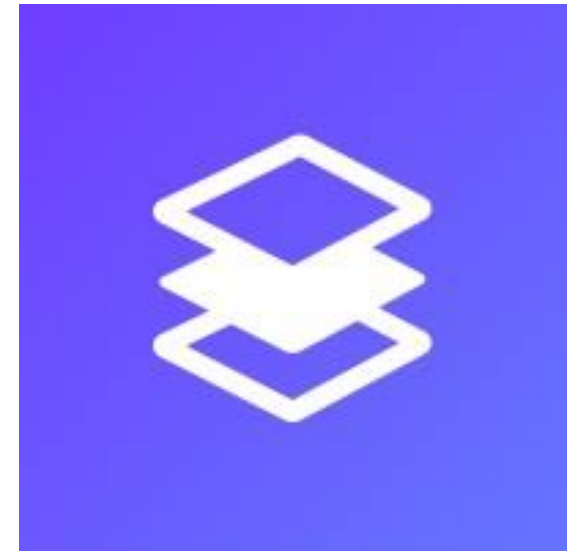
| Feature Name            | Feature Description                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Block Difficulty        | Unitless, relative measure of computational power required to mine a new block on the blockchain.                                                  |
| Block Mint Reward       | The reward (typically in the respective cryptocurrency) for successfully mining a new block.                                                       |
| Block Stripped Size     | The size of a given block on the blockchain (in bytes) stripped of signature bytes (included for validation and do not include relevant metadata). |
| Total (Gas) Fees        | Fees collected by miners to host transactions on a given block.                                                                                    |
| Block Transaction Count | The number of transactions which have occurred on a given block.                                                                                   |
| Block Weight            | A measure of a given block's size and complexity, weighing in favor of blocks with more transactions on them.                                      |

# Data Sources

- Hourly (aggregated) On-Chain data was pulled from [Dune](#)
- Hourly Price data was pulled from [CoinAPI](#)
- Staking Data was pulled from [StakingRewards](#)



Staking Rewards



# Strategy Overview

- Our strategy aims to leverage untransformed crypto data alongside constructed technical indicators (Fibonacci retracement, RSI, EMA, etc.) to forecast strategy moves (buy, hold, sell).
- These indicators will form the data inputs for a Neural Network (Multi-layer Perceptron; MLP) model, aiming to identify profitable patterns and weed out irrelevant features.



# Parameterization

The model itself features the following parameters:

- $b$  := lookback window size
- $f$  := prediction horizon
- $x$  := lower prediction threshold
- $y$  := upper prediction threshold

# Segmentation

- Since crypto price data has significant time-varying volatility and dynamics that can negatively affect model training, we use a change point detection algorithm called PELT to mitigate these issues
- PELT allows us to identify "change" points in the time-series data, which creates several segments with their own unique statistical dynamics
- Using segmentation, we utilize a more robust methodology for normalization of non-stationary time-series data

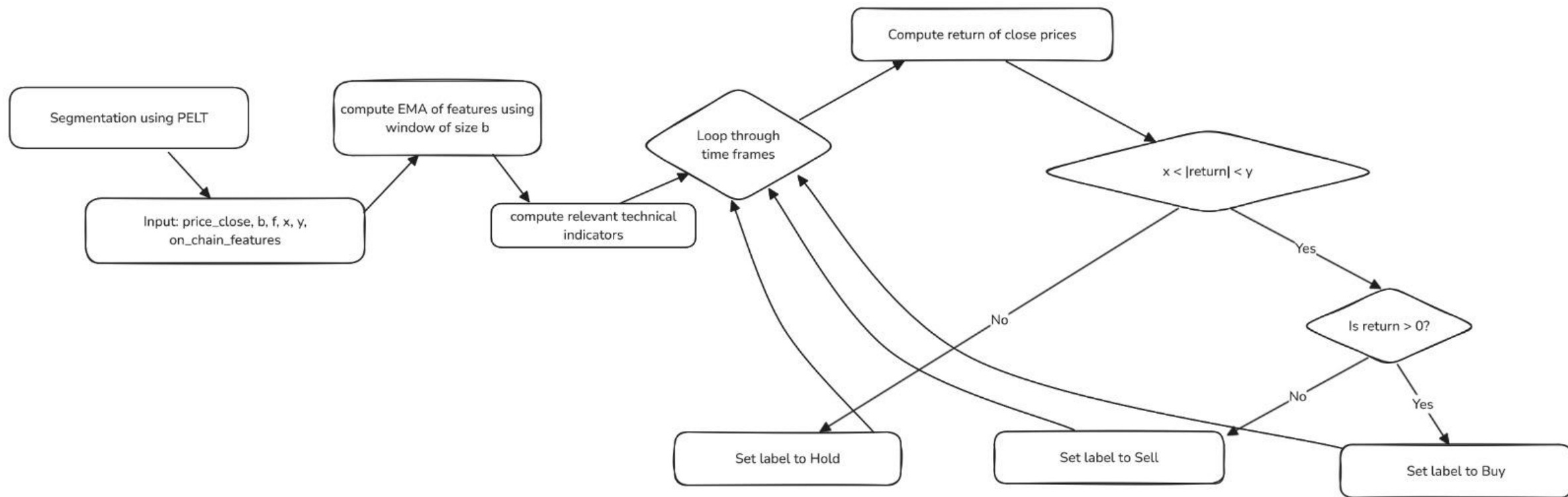
# Model Construction

Our model is a 4-layer Multi-layer Perceptron (MLP).

- The input layer contains all the data discussed in our feature space
- The output layer consists of three nodes (buy / sell / hold) which will be interpreted as our position actions.

The two hidden middle layers at present have an unspecified number of nodes, but this will be a parameter that we can fine-tune as we test our strategy's performance.

# Model Pipeline



# Backtesting - Assumptions

For our backtesting, we plan to include the following parameters:

- $f$  := trading fees of 10 bps
- $l$  := stop loss threshold of 1% of portfolio
- $K$  := initial portfolio capital of 100,000
- This gives us the flexibility to closely replicate actual trading conditions and implement safeguards that may optimize performance in out-of-sample performance.
- Our movements have no market impact (prices are not retroactively changed due to our trades)
- We can make trades at closing price at the end of each hour with no slippage

# Portfolio Allocation

Our portfolio allocation strategy takes advantage of the probabilities output by the softmax activation function.

We assume the equilibrium assumption and assign each token 20% of the initial starting capital as its investment pool. When we receive a 'BUY' signal, we take the corresponding softmax probability associated with this classification and treat it as a *strength* indicator. We size the position by taking the strength indicator divided by some constant  $C$  to get some proportion  $p$  of our maximum capital allocation which is used for the position.

We perform a symmetrical procedure when we receive a 'SELL' signal. 'HOLD' signals lead to no changes in the portfolio holdings.

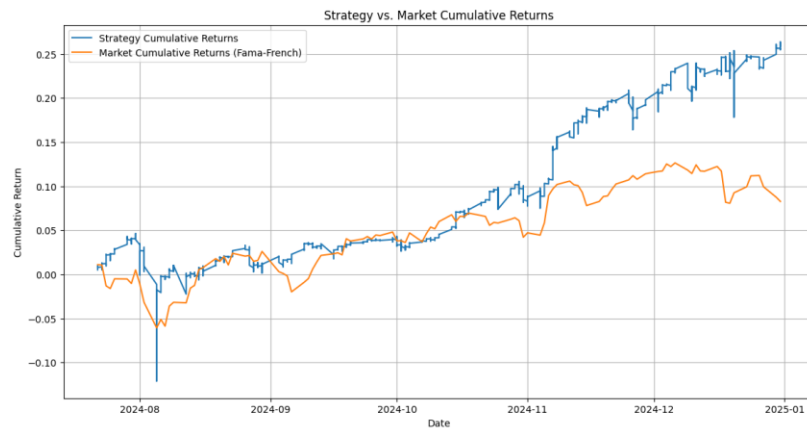


# Analysis AVAX

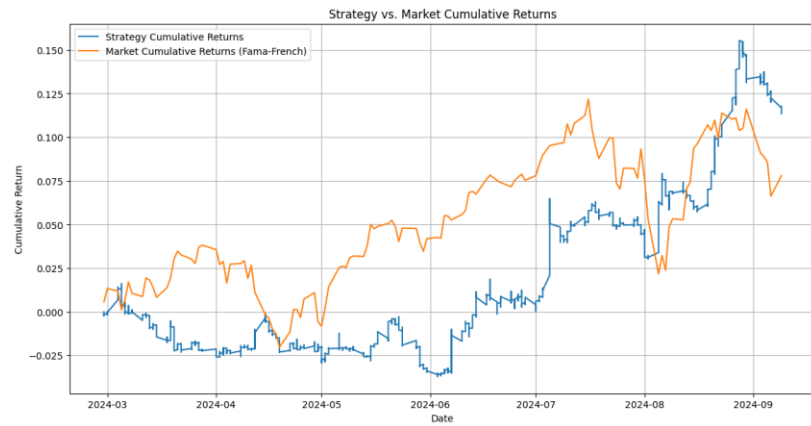


- Annual Returns: 0.38
- Sharpe Ratio: 1.76
- Max Drawdown: -0.115
- 5% VaR: 0.002
- Beta to BTC: 0.14

# Analysis SOL



- Annual Returns: 0.04
- Sharpe Ratio: 0.12
- Max Drawdown: -0.25
- 5% VaR: 0.004
- Beta to BTC: 0.30



# Analysis MATIC

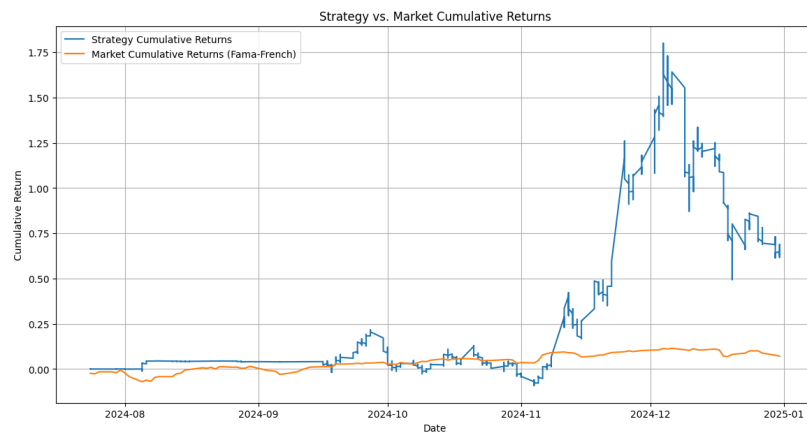
- Annual Returns: 0.22
- Sharpe Ratio: 1.95
- Max Drawdown: -0.05
- 5% VaR: 0.001
- Beta to BTC: -0.03

# Analysis NEAR



- Annual Returns: 0.23
- Sharpe Ratio: 1.77
- Max Drawdown: -0.083
- 5% VaR: 0.002
- Beta to BTC: -0.09

# Analysis DOT

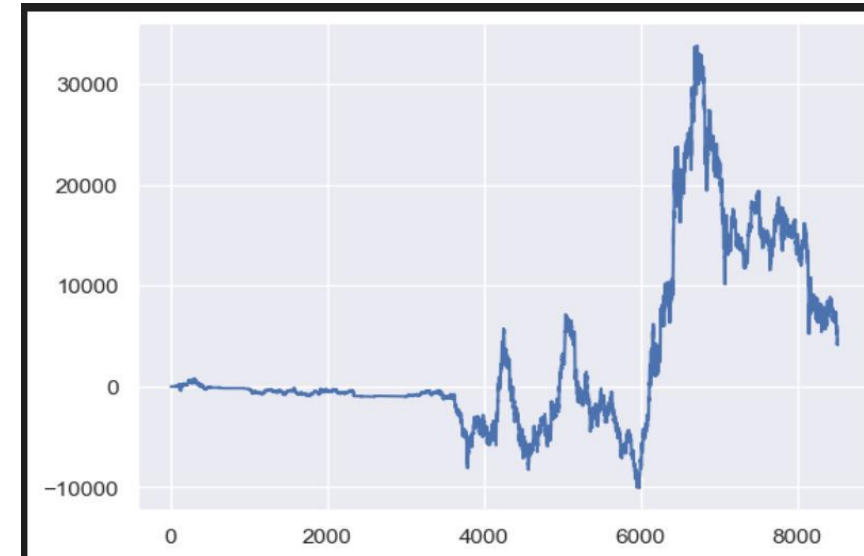


- Annual Returns: 0.64
- Sharpe Ratio: 0.69
- Max Drawdown: -0.64
- 5% VaR: 0.015
- Beta to BTC: 0.95

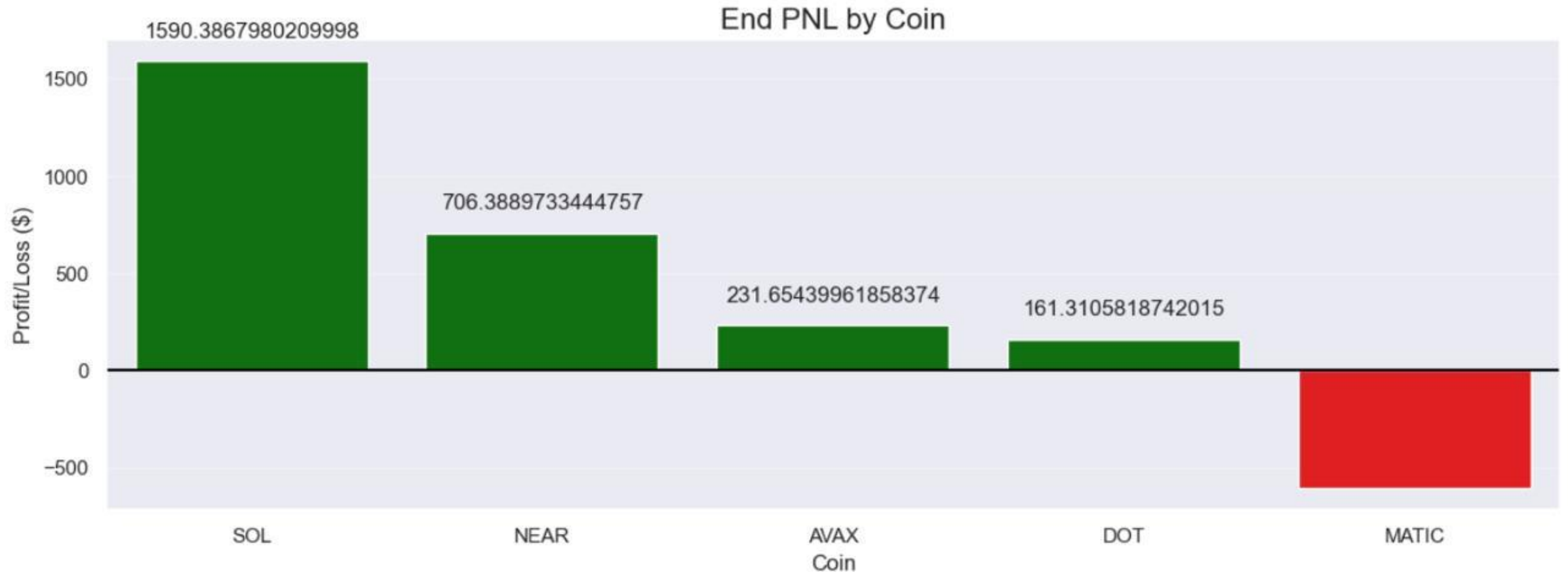
# Portfolio Results (Performance Metrics)

- Stop loss allows us to shift our focus to the coin that we're predicting more accurately.
- Graph 1 depicts portfolio performance metrics PnL
- Graph 2 depicts portfolio PnL over time

| Performance Metrics |       |
|---------------------|-------|
| mean_return_annual  | 0.08  |
| vol_annual          | 0.26  |
| sharpe_ratio        | 0.31  |
| sortino_ratio       | 0.36  |
| max_drawdown        | -0.22 |
| calmar_ratio        | 0.35  |
| VaR_95              | 0.00  |
| CVaR_95             | 0.01  |



# PNL by Coin





# Portfolio Risk Management

- Implemented an extremely tight stop loss at 1%
- Equal allocation to position sizing to eliminate over exposure to any single asset
- Each trade is scaled according to the probability weights based on the MLP
- Higher probability leads to larger position sizes

# Algorithm Launch

- Assumed Capital \$100,000
- Equally Weighted across 5 tokens (\$20,000)
- We trained each model across each token
- Training time ~ 90 seconds
- Buy and sell signals are based  $\frac{1}{2}$  of the standard deviation of the change in price-close

# Conclusion

- Thank you for taking the time to view our algorithm. Please refer to the technical paper for an in-depth analysis on our model and rationale.

# Sources

- Kim, G., Shin, D.-H., Choi, J. G., & Lim, S. (2022). A deep learning-based *cryptocurrency price prediction model that uses on-chain data*. IEEE Access, 10, 56232–56247. <https://doi.org/10.1109/ACCESS.2022.3177888>
- Parente, M., Rizzuti, L., & Trerotola, M. (2024). A profitable trading algorithm for cryptocurrencies using a neural network model. Expert Systems with Applications, 238, 121806. <https://doi.org/10.1016/j.eswa.2023.121806>